

A Feature-Based Corpus Typology of Perfect-Like Grams

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1 Introduction & Motivation

1.1 Project Overview

Perfect-like grams (PLGs) show systematic cross-linguistic patterns shaped by morphology-semantics mapping, language-specific TAM systems, and potentially other interacting factors. This project seeks to develop a feature-based typological model using parallel corpora and computational pipelines to explore cross-linguistic correlations among these factors. To test this framework, I use machine learning classifiers to predict PLGs across languages. Their outputs are then analyzed with inferential statistics, weighted error diagnostics, and feature-based semantic mapping, producing results that are both computationally rigorous and linguistically interpretable. While the methodology has a computational backbone (data science, NLP, machine learning), the ultimate goal is linguistic: refining our understanding of PLGs, highlighting underdescribed variation, and connecting computational tools back to linguistic theory.

This dissertation is comprised of three interlocking projects. Together they (i) evaluate the ability of an ontological feature schema to capture cross-linguistic distributions and interpretations of PLGs, including comparison with vector-based distributional models such as BERT and S-BERT; (ii) uncover novel insights into the forms and meanings of PLGs and into typological comparison more broadly; and (iii) test the scalability of this approach to generalize to new languages and unseen data.

1.2 Aims & Research Questions

The project pursues four mutually reinforcing aims:

1. **Linguistic insight.** The core aim of the thesis is to leverage computational methodologies to better understand and explain linguistic phenomena. The object of study, cross-linguistic perfect-like grams (PLGs), is a rich domain for this undertaking as it sits at the junction of the theory and typology tense-aspect-mood (TAM) research. This project focuses on using classification experiments alongside semantic mapping tools to test how well a theory- and typology-informed ontological feature schema can capture cross-linguistic patterns in PLGs through observing both the predictive capabilities of such a linguistically-grounded schema as well as clustering patterns of meanings, morphological expression, language-specific TAM system, and other factors that may be at play. This will be accomplished by ensuring that the computational framework remains linguistically informed, while also generating new evidence to inform debates in TAM typology, language description, and theory.
2. **Feature schema development.** Central to the project is the gram and iterative refinement of a cross-linguistic PLG feature inventory, grounded in TAM literature and corpus evidence,

with diagnostics for interpreting feature values. Corpora such as the Bible and TED Talks will provide initial data for testing and refinement, and scalability testing will incorporate data from the DoReCo corpora.

3. **Cross-disciplinary integration.** This project further seeks to demonstrate how computational pipelines can both sharpen and address linguistic questions while linguistic theory can enrich computational models, producing a roadmap for ongoing, cross-disciplinary collaboration between linguistics and NLP.
4. **Ontological innovation.** Ultimately, the goal of this research program is to develop a prototype for applying ontology-driven schemas to other domains (e.g. clause type contrasts, TAM ambiguity, low-resource languages). This involves piloting an ontological feature schema in machine learning, encoding semantic featural distinctions such as experiential versus resultative readings into a structured, machine-readable ontology. This allows model performance to be interpreted in typologically and semantically meaningful terms, establishing a pathway to future scalability and boarder applications.

These aims coalesce into two central questions, which guide the project as a whole.

- To what extent can a feature-based semantic ontology, developed on the basis of aligned corpus data, but annotated on a single language, model cross-linguistic variation in PLGs?
- Does the application of such a schema reveal typologically consistent contrasts (e.g. experientiality, evidentiality, so-called “current relevance”) across languages that differ morphosyntactically and genealogically, providing a scalable framework for typological comparison, semantic annotation, and temporally complex meanings?

Concomitant operational questions include:

- Is the schema internally coherent and usable across languages?
- Can the features be clearly defined and consistently annotated?
- Do these features effectively capture the typological diversity of PLGs across languages?
- How are these features instantiated (morphologically/morphosyntactically) across languages, what systematic profiles emerge, and what correlations can be drawn between these factors?
- Are there recurring patterns or shared strategies across unrelated languages?

Together, these questions position the project at the intersection of typology, description, and computation, asking not only whether PLGs form a coherent cross-linguistic class (cf. Dahl & Wälchli 2016), but also whether a feature-based approach can make that coherence (or lack thereof) explicit, interpretable, and empirically testable.

1.3 Background & Literature Review

1.3.1 Perfect-Like grams: A Cross-Linguistic Puzzle

Perfect-like grams (PLGs) sit at the intersection of tense, aspect, modality (TAM), and information structure, yet resist tidy definition even within individual languages, let alone across them. While many languages possess TAM categories that have been compared to the English

present perfect (e.g. French *passé composé*, German *Perfekt*, Turkish evidentials, Mandarin *guò*), the semantic contributions of these forms vary widely, often defying one-to-one translation and challenging conventional comparative frameworks (Bertrand et al., 2022; Dahl, 2000; Dahl & Wälchli, 2016).

Despite their apparent shared characteristics, such as links to past events, non-narrative temporal contours, or present relevance (“current relevance”, “discourse salience”, etc.), PLGs have long been recognized as resistant to neat categorizations (Plungian, 2011; Schaden, 2009), a quality reflected in the ongoing theoretical debate over their core semantics. For example, even closely related European languages diverge sharply in how PLGs interact with tense and aspect systems. As shown in (1a, b), in French and German, PLGs are compatible with definite temporal adverbial modification (“DTAM”), while in English and Spanish (examples (1c, d)) they are not¹.

- (1) a. English
I have read this book (*yesterday).
- b. Spanish
He leído este libro (*ayer)
have.1SG.PRES read.PTCP this book (*yesterday)
‘I have read this book (*yesterday).’
- c. German
Ich habe diese Buch gestern gelesen
I have.1SG.PRES this book yesterday read.PTCP
‘I [have] read this book yesterday.’
- d. French
J’ai lu ce livre hier
I.have1SG.PRES read.PTCP this book yesterday
‘I [have] read this book yesterday.’

Such contrasts show how easily even typologically close languages break apart under fine-grained semantic diagnostics, highlighting the limits of categorical comparison.

Even within a single language, multiple PLG strategies can coexist, some overlapping with English perfects and others diverging in subtle but systematic ways. In Japanese for instance, as shown in (2), [verb + *koto ga aru*] expresses an experiential reading similar to English perfects, while the simple past *-ta* form can yield resultative interpretations in certain contexts, as shown in (2b) below². Georgian likewise illustrates this point: the Synthetic Perfect carries evidential force, while the Periphrastic Perfect does not, such that (3a) is only felicitous in contexts where the speaker wishes to convey a particular stance with respect to the source of knowledge (in the Georgian Synthetic Perfect, visual evidence or second hand – including traditional – knowledge), where as (3b) is evidentially neutral, and must explicit include an adverbial (such as *turme* ‘apparently’ or *rogorts chans* ‘it seems that’) in order to convey a similar meaning³.

¹This contrast is widely discussed in the literature as the “Present Perfect Puzzle” (e.g. Klein 1992; Iatridou, Anagnostopoulou, & Izvorski 2001; Pancheva 2004; many others).

²Sakai 1981; Ogihara 1989

³Makharoblidze (2012); Hewitt (1996); many others

- (2) a. Experiential *koto ga aru*
 nihon ni itta koto ga aru
 Japan to go.PAST NMLZ PTCL be
 ‘I have been to Japan.’ (Lit. ‘there is an event in which I went to Japan’)
- b. Resultative *-ta*
 tsukareta
 tire.PAST
 ‘I am tired.’ (Lit. ‘I [became] tired’, ‘I [have become] tired’)
- (3) Georgian Synthetic and Periphrastic Perfects
- a. Synthetic Perfect with grammatical evidentiality
 Anzor-s es ts’igni ts’a-u-k’itkh-av-s
 Anzor-DAT this book PV-VS-read-TS-3SG
 ‘Anzor has apparently read this book.’
- b. Periphrastic perfect with neutral evidentiality
 Turme / rogorts chans Anzor-s es ts’igni ts’a-k’itkh-ul-i akvs
 apparently / it seems that Anzor-DAT this book PV-read-PTCP-NOM have.3SG.PRES
 ‘Apparently/it seems that Anzor has read this book.’

The DTAM contrast illustrates how PLGs diverge across closely related languages, while the Japanese and Georgian examples show how multiple PLG strategies can coexist within a single language. Together, these patterns highlight the need for methods that go beyond categorical comparison and instead track underlying features. By moving from categorical comparisons to feature-based analysis across large, multilingual corpora, this project aims to reveal cross-linguistic contrasts and commonalities that would be invisible in small-scale elicitation or in one-to-one translation equivalence. This typological diversity, both across and within languages, is precisely what makes PLGs a compelling object of typological and computational inquiry: their semantic subtlety and morphosyntactic variability pose significant challenges for formal theory, while offering a valuable testing ground for empirically grounded, feature-driven models.

To situate these puzzles within broader debates, we turn to the core readings canonically attributed to the perfect: experiential (emphasizing that an event has occurred at least once before the speech time), resultative (focusing on a present state that results from a past event), universal (a state or event beginning prior to speech time and continuing up to and/or through it), and recent past (a situation occurring shortly before speech time). Each of these readings is exemplified in (4)

- (4) a. **Experiential:** She has read this book (before / three times / at some point).
 b. **Resultative:** The sun has set (and now it is dark).
 c. **Universal:** We have lived here (for three years / since 2007).
 d. **Recent Past:** The letter has just arrived.

1.3.2 Theoretical Foundations: Core Semantic Readings of the Perfect

As stated above, PLGs have typically been associated with a small set of recurring interpretations. These readings, while not always discrete in practice, have provided a useful descriptive

framework for comparing TAM grams across languages (Comrie, 1976; Klein, 1992; McCoard, 1976; Mittwoch, 2008). As seen in studies such as Dahl (1985) and Plungian & Auwera (2006), these categories can be used to distinguish PLGs across genealogically diverse languages.

In the context of cross-linguistic PLG comparison, experientiality offers a testable feature, namely, the presence of quantified event recurrence such as in *I have eaten durin [before, three times, etc.]*, that can guide corpus annotation and comparison. Resultatives, by contrast, imply a change-of-state event whose resultant state holds at reference time (e.g. the window has broken, the butter has melted). Universals extend the predicate across a continuous time span up to the reference point (e.g. I have lived here for five years, she has always had brown hair). While these categories reflect theoretical abstraction, they remain widely used in typological and computational work, e.g. perfect-aspect tag distinctions in the Universal Dependencies project or in multilingual NLP tasks like temporal relation extraction (Zhao 2022a).

Departing from traditional typological treatments, the current project takes these readings not as mutually exclusive categories, but rather as semantic features that potentially co-occur. For instance, a single gram may simultaneously express experiential and resultative features, a pattern that resists categorization in existing discrete typologies but aligns with a feature-based representation. This approach reflects a move away from discrete categories toward feature-based modeling approaches (Bertrand et al., 2022; Haspelmath, 2010a,b).

1.3.3 Cross-Linguistic Approaches: From Nebulous Categories to Predictive Features

The typological complexity of PLGs has been extensively documented in survey work and large-scale comparative projects. Seminal work by Dahl (2000) and Dahl & Velupillai (2013) established broad patterns in TAM systems across languages, while more recent studies (Bertrand et al., 2022; Bowler & Ozkan, 2017; Zhao, 2022a,b) have emphasized the gradient, overlapping nature of the semantic denotations of PLG-like grams (e.g. perfect-like grams in Turkish and Bulgarian encode both anteriority and evidentiality, but differ in source and speaker orientation⁴) across genetically and areally diverse systems.

A key insight from this literature is that cross-linguistic variation is often more a matter of degree than kind, which is to say that many languages encode 'perfect-like' meanings, such as recurrence, relevance to reference time, or resultant state, using tense-aspect combinations, adverbials, or particles, rather than grammaticalizing a distinct perfect gram. The overlap or convergence between perfects and iamitives (Dahl & Velupillai 2013), as well as the often fuzzy boundary between resultative and evidential readings (Plungian 2011), reinforce the need for modeling strategies that represent cross-linguistic polysemy without reducing diverse grams to a single cross-linguistic category.

This has led to a recent wave of work attempting to identify and formalize cross-linguistically predictive features of TAM categories (e.g. iterative experientiality, persistent state, or temporal anchoring) from corpus data and annotated language resources, a trend this project builds upon to enable more precise and nuanced cross-linguistic comparison and linguistic data annotation. Of particular relevance is the work of van der Klis et al. (2017, 2021), LeBruyn et al. (2019), and Zhao (2022, 2022), who propose annotation schemas and classification models that incorporate semantic, pragmatic, and discourse-related factors. However, these models often rely on fixed categorical inventories of TAM labels (e.g. Resultative, Perfective), rather than decomposing grams into cross-linguistic semantic features and studying how those features cluster. This project diverges by embracing a scalable, feature-oriented ontology informed by cross-linguistic corpus data, which

⁴Bowler & Ozkan (2017)

organizes TAM grams through cross-linguistically predictive semantic features and their clustering patterns, rather than through fixed categorical labels.

1.3.4 Methodological Orientation: Toward Semantics-Lite Ontologies

The goal of this project is to develop a usable, interpretable feature set that can track variation in PLGs across typologically distinct systems, and that can be productively deployed in both research and applied NLP contexts, facilitating dialogue between these domains. While indebted to formal semantic accounts of the perfect (Portner 2003, 2011; Kratzer 2004; Iatridou, Anagnostopoulou, & Izvorski 2001), the project is not primarily concerned with extending that tradition, but rather with building a practical feature ontology for cross-linguistic comparison and annotation.

In doing so, the project draws a working distinction between inherent semantic features, i.e. those conventionally encoded in the truth conditional semantics of a gram⁵, and context-dependent interpretations, which may arise from pragmatic inference or usage context. This distinction, while not exhaustively formalized in the pilot, informs both feature design and annotation logic, and reflects ongoing conversations in formal semantics around the boundary between encoded and inferred meaning.

This approach aligns with recent calls in linguistic typology and computational semantics for hybrid models, ones that acknowledge both the utility of comparative concepts and the importance of language-specific, gram-based patterns (Haspelmath, 2010a,b, 2016). By grounding feature definitions in observable diagnostics (e.g. compatibility with temporal adverbs, stative versus eventive predicates, etc.), and by testing them against real corpus data, this work aims to strike a balance between cross-linguistic abstraction and alignment with attested usage patterns across languages. In doing so, it hopes to extend the toolkit available for modeling temporally complex meaning in natural language, while also providing a more interpretable alternative to opaque, large-model-derived representations (cf. Chen et al. 2021).

2 Pilot Experiment

This section presents the completed pilot experiment which informs and shapes the following work of the project.

2.1 Motivation & Goals

Perfect-like grams (PLGs) pose challenges for both linguistic typology and computational modeling. Their semantic and morphosyntactic complexity (such as the interaction of anteriority with evidentiality, or variation in auxiliary selection across contexts) makes them difficult to capture through traditional cross-linguistic comparison, while their variability (such as context-sensitive shifts between experiential and resultative readings) also poses difficulties for current NLP approaches, which often rely on surface-level cues or discrete category labels in tasks like tense detection, temporal relation classification, and cross-lingual parsing. These systems may misinterpret perfect-like forms as simple pasts, or fail to distinguish between readings that differ subtly in temporal or evidential nuance.

The three principle aims of the pilot were to assess whether (i) the proposed schema, a feature-based annotation framework for PLGs (see §2.2.1) could be consistently applied across languages;

⁵E.g. a gram may inherently encode anteriority, but be interpreted as evidential or experiential depending on discourse context.

(ii) to test whether classifiers could reliably learn mappings between feature profiles and grams; and (iii) to evaluate whether the resulting errors revealed meaningful feature-patterns for analysis.

As the first implementation of the schema, the pilot provides a foundation for scaling to more languages and refining the schema iteratively. The insights gained here inform methodological frameworks for subsequent experiments, including empirically grounded adjustments to feature design and annotation strategies.

This work builds on research in typological approaches to TAM (e.g. Bertrand et al., 2022; Le Bruyn, van der Klis, & de Swart, 2019; van der Klis, Le Bruyn, & de Swart, 2021; Zhao, 2022a,b) and computational studies of error analysis (Belinkov & Glass, 2019; Geva, Goldberg, & Berant, 2019; Ribeiro, Singh, & Guestrin, 2016). It extends both by applying a novel feature-based methodology to cross-linguistic corpus data. The following sections detail the methodology and results of the pilot study.

2.2 Methodology

The pilot experiment used a small verse-aligned corpus of New Testament Bible (NTB) texts in English, German, and Mandarin. A total of 300 tokens were extracted, each representing a distinct instance of a perfect-like gram (PLG). Tokens were identified in the English NTB using an automated detection method (spaCy) that locates all instances of auxiliary *have* in co-occurrence with a verbal noun (as defined by syntactic structure and typical morphological cues).

Because this extraction method is anchored in English (*have* + participle), it risks bias toward forms overtly marked in English. To mitigate this, subsequent experiments will explore reverse seeding techniques. In this approach, English PLGs are first aligned with non-English texts, and the grams that most consistently co-occur with English perfects (e.g. the German Perfekt, Mandarin *le*, Georgian *turmeobiti*) are identified. These grams can then be used as search terms in the reverse direction, retrieving all their instances in non-English texts and examining the corresponding English alignments. Conceptually straightforward, reverse seeding is technically nontrivial: alignment noise, multifunctional markers, and the need for statistical thresholds (e.g. association strength, coverage) require careful filtering for reliable results. Addressing these challenges is part of the focus of the project’s first formal phase in Fall 2025.

Once extracted, the data are manually annotated⁶. The NTB corpus is pre-aligned at the verse level, with word-level alignment available to varying degrees across languages. While verse alignment enables reliable cross-linguistic comparison of grams in context, word alignment quality varies considerably across language pairs; alignments are strong for German, but considerably noisier for others like Aymara and Xhosa.

A crucial distinction exists between inherent semantic features (properties encoded by the gram itself) and contextual/construal features (interpretations that arise in use). While the current pilot does not attempt a systematic separation of these layers, ambiguous cases where context appears to override inherent encoding are flagged in the metadata for reference in error analysis. A fuller implementation of this distinction is an important direction for future work.

Finally, the pilot featural annotation and gram identification were conducted by a single L1 English speaker, viz. the author. This poses two limitations. First, in the featural annotation, certain nuances in the non-English texts may not be fully captured. Second, in the gram-identification annotation, some non-English grams may be misidentified or overlooked. Subsequent studies will

⁶In this project, “annotation” refers to two processes: (i) coding each PLG instance for semantic features, and (ii) identifying which gram is used in each language (e.g., the German Plusquamperfekt or the Mandarin particle *guò*). Feature coding uses the English text as an “anchor,” assuming that aligned grams encode equivalent features via translational correspondence.

address these limitations by recruiting L1 research assistants to conduct the non-English gram identification and, where relevant, to support featural annotation by flagging cases in which the non-English text encodes features absent in the English. This work will be guided by a clear annotation manual and checked via inter-annotator agreement (IAA).

2.2.1 Features

The features used in the pilot experiment are all drawn directly from the formal semantic literature on perfects, and represent some of the most commonly cited meaning components associated with PLGs cross-linguistically. Each feature used in the pilot experiment is presented here along with the operationalized definition and diagnostic(s) used to evaluate its presence or absence in each individual token. Further discussion of the performance and impact of performance on feature scheme refinement is discussed in depth in the ?? section below.

2.2.1.1 Experiential Reading

One of the “core” readings of the perfect is the experiential, which indicates that subject of the predicate has the experience of performing or undergoing the event at least once within a time span that is anterior to the reference time (or Topic Time) established by the perfect gram (Klein 1992). This includes expressions like *she has read this book before*, and *Padma has visited Bangkok many times*. The operationalized definition of this feature which is used in the pilot study is as follows:

A gram expresses an experiential reading if it conveys that at least one instance of the eventuality described by the predicate has occurred prior to the reference time, without necessarily implying an ongoing result state.

Because both the experiential and resultative readings fall under the umbrella term “existential”, as both readings involve existential quantification, clear diagnostics are needed to distinguish them⁷. The first diagnostic makes use of frequency or experience markers, such as *never*, *often*, and *ever* (Mittwoch 2008, Ritz 2012): *I have often visited that park*, *she has never read that book*, and *they have been to Paris three times* all exemplify this diagnostic. Similarly, experientials are incompatible with modifiers that target result states, as exemplified in (5) below.

- (5) a. I have been to Tokyo, and therefore am familiar with its metro system. (experiential only)
- b. I have eaten the rice, so now it’s gone. (resultative only)

Kiparsky (2002) and Ritz (2012) further observe that experiential readings are generally infelicitous with *wh*-questions unless the question targets the event itself, rather than a resulting state. For example, in (6a) where targets the event of hiding, whereas in (6b) it targets the present location resulting from the hiding event, rather than the event itself.

- (6) a. Where have you found my watch? (experiential only)
- b. Where have you hidden my watch? (slightly ambiguous but strongly favors resultative)

⁷ In cases where grams were genuinely ambiguous between experiential and resultative readings (approximately 5% of 300 pilot tokens), both features were marked as positive. These instances were tagged in the metadata to allow for downstream filtering or exclusion during analysis.

In the pilot, grams were annotated as experiential if they expressed a past event without an explicit result state, and were compatible with markers of recurrence, quantity, or narrative summary (e.g. *many times*, *before*, etc.).

2.2.1.2 Resultative Reading

Another “core” reading of the perfect is the resultative, which indicates that the occurrence of a change-of-state event anterior to the reference time instantiates a result state that holds at that reference time (Alexiadou, Rathert, & von Stechow, 2003; Partee, 1973). This includes expressions like *he has broken his arm [and it is now broken]* and *they have eaten all the cake [so now there is none left]*. The operationalized definition of this feature which is used in the pilot study is as follows:

A gram expresses a resultative reading if it encodes or implies a state that results from a change-of-state event, and that that state holds up to, and potentially includes, the reference time (e.g., the speech time in the case of present perfects).

Several of the diagnostics used above to disambiguate experientials from resultatives can be used in reverse to disambiguate resultatives from experientials⁸. For example, resultatives are compatible with result state modifiers, as shown in (7) below (Alexiadou, Rathert, & von Stechow, 2003; Kratzer, 2000; Partee, 1973).

- (7) a. I have broken the vase, and it’s in pieces (resultative only)
- b. He has read that book [before], #and now it’s in the state of having been read (result state nonsensical)

Likewise, resultatives are not felicitous with frequency or experience markers, leading to the distinction seen in (8).

- (8) a. I have sprained my ankle twice before. (does not entail current result state; experiential only)
- b. #I have often broken my leg. (experiential only)
- c. I have broken my leg [and it is now broken]. (resultative)

An additional diagnostic is the incompatibility of resultatives with *for* adverbials, which distinguishes resultatives from universals, which are felicitous with *for* adverbials, as seen in (9).

- (9) a. I have lived here for five years. (universal and/or stative)
- b. *I have broken my leg for five years. (ungrammatical)

In the pilot, grams were annotated as resultative if they entailed a state persisting from a change-of-state event and were compatible with modifiers indicating present relevance or result.

⁸See footnote 7.

2.2.1.3 Universal Reading

The last of the “core” perfect readings included in this feature set⁹ is the universal, which unlike the experiential and resultative use of existential quantification instead operates on universal quantification over a time span (hence the name), which is often but not exclusively identified overtly in the context. This includes expressions like *I have always lived here, she has had blue hair since last month, and they have been reading for three hours*. The operationalized definition of this feature which is used in the pilot study is as follows:

A gram expresses a universal reading if it conveys that a state or event holds continuously throughout a contextually defined time span, typically up to (but not necessarily including) the reference time.

Perhaps the most commonly used diagnostic for identifying universal perfects makes use of their compatibility with *for* adverbials (Comrie, 1976; Iatridou 2003), as exemplified in (10).

- (10) a. She has studied Latin for five years. (*for* adverbial felicitous $\rightarrow$ universal)
 b. #He has broken his leg for five years. (non-durable event incompatible with *for* adverbial)

Another prominent characteristic of universal perfects is their durative predicate class restrictions: universals are only felicitous with states and activities, i.e. atelic/durative predicates (Comrie, 1976; Dowty, 1979). This is exemplified in (11).

- (11) a. I have always been tall. (state $\rightarrow$ universal available)
 b. She has been swimming for an hour. (activity $\rightarrow$ universal available)
 c. #They have painted the house for seven years. (universal available only if interpreted as habitual activity)
 d. #He has arrived for three days. (achievement $\rightarrow$ universal unavailable)

Thirdly, (English) universals allow for paraphrasing of *since* adverbials with *all along*¹⁰ (Comrie, 1976; Iatridou 2003), as shown in (12).

- (12) a. She has known Anto *since childhood*. $\rightarrow$ She has known Anto *all along*. (universal available)
 b. She has visited Anto *since college*. $\nrightarrow$ She has visited Anto *all along*. (iterative, not universal)

It has also been noted (Dahl & Velupillai, 2013; Plungian & Auwera, 2006; Schaden, 2009) that many languages do not encode universal readings via PLG at all, encoding universal-type readings present tenses, generally accompanied by progressive or habitual aspect.

In the pilot annotation, universal readings were identified where the predicate described a state or activity holding continuously over a contextually bounded interval (Comrie, 1976; Iatridou 2003). These readings were restricted to durative predicates and were often difficult to distinguish from stative presents in translation.

⁹The fourth is the “recent past” reading, which is generally treated as a pragmatic nuance of the experiential and resultative readings. While occasionally marked morphologically in some languages, in most systems it is more often inferred from context and adverbial modification, such as *just* (Comrie, 1976; Dahl, 1985; Kiparsky, 2002; Iatridou 2003).

¹⁰The specific temporal anchor is lost, but the continuity or persistence of the event (“temporal contour”) remains available through contextual inference.

2.2.1.4 Compatibility with DTAM

This feature refers to the ability of a present-situated PLG to co-occur felicitously with (past referencing) definite temporal adverbial modification (DTAM), such as *yesterday*, *in 1999*, or *last week*; i.e. time expressions which refer to a specific, bounded point or interval in the past. The operationalized definition of this feature which is used in the pilot study is as follows:

A token is annotated as positive for DTAM if it permits felicitous co-occurrence with such adverbials.

This feature is of particular interest in the study of PLGs due to the well-documented “Present Perfect Puzzle” (Klein 1992; Pancheva 2004; Pancheva & von Stechow 2004; many others), i.e. the differing language-specific compatibility of present-situated PLGs with DTAM. This is exemplified below in English (13a), Spanish (13b), German (13c), and Serbian/Croatian (13d)¹¹.

- (13) a. You heard it yesterday (*have heard).
 b. Ya lo oísteis ayer (*habéis oído).
 already it hear.2pl.PRET yesterday (*have.2pl.PRES hear.PTCP)
 ‘You heard it yesterday (*have heard)’.
 c. Ihr hat es gestern gehört.
 you have.2pl.[PRES] it yesterday hear.PTCP
 ‘You (have) heard it yesterday’.
 d. To ste juče čuli.
 that you yesterday hear.PERF
 ‘You (have) heard it yesterday’.

In the pilot dataset, present perfects in English were uniformly annotated as negative for DTAM due to the nature of the extraction process. This bias will be addressed in future iterations via reverse-seeding methodologies, as described in the introduction to this section.

2.2.1.5 Evidentiality

Evidentiality refers to linguistic strategies for conveying the source of information or knowledge conveyed in a given utterance. In some languages grammatical systems, evidentiality is overtly encoded, such as via a dedicated morpheme or truth conditional semantics of certain grams more generally, while in others (such as English), it is expressed by means of circumlocutions such as *apparently, I hear (that...)*, or *they told me (that...)*. The operationalized definition of this feature which is used in the pilot study is as follows:

A gram expresses evidentiality if it overtly marks the source of information, either through grammatical markers or lexical/discourse framing (e.g. *apparently, I heard, they say*).

The examples in (14) demonstrate evidentiality in English, which does not grammaticalize source of information.

¹¹Umberto Eco, ParaSol Corpus, via Grønn & von Stechow (2021).

- (14) a. He has eaten all the cake.
 b. Apparently he has eaten all the cake.
 c. I heard he has eaten all the cake.

In (14), though English does not grammatically encode evidentiality, each expression conveys a differing attitude of the speaker toward the information conveyed (including, pragmatically, degree of confidence in the information). In (14a) the information is conveyed as epistemically neutral; no source of information is specified, and the assumption is that the information is factual. In (14b) while no specific source of information is expressed, the use of *apparently* provides the speaker with a certain amount of confidence “wiggle room”; the information is plausible, even probable, but the speaker is not committing to it entirely, i.e. epistemic hedging. In (14c) the source of information is not expressly named, the speaker is expressly placing confidence in the information on an outside source.

Next consider the examples in (15), from Georgian, a language which does grammatically encode evidentiality.

- (15) a. mteli namtskhvari sheuch'amia.
 all cake eat.TUR-I
 ‘He has (apparently) eaten all the cake’.
- b. mteli namtskhvari shech'muli akvs.
 all cake eat.PTCP have.3SG
 ‘He has eaten all the cake’.

Both examples in (15) can be faithfully translated in English as *he has eaten all the cake*, though each conveys a different evidential meaning. (15a) uses the Synthetic Perfect (*turmeobiti I*), which intrinsically conveys an evidential stance such that the information expressed is gleaned either through hearsay or deduced from visual evidence. This expression is only felicitous in contexts where one or both of these factors is present. By contrast, (15b) uses the periphrastic perfect, which is epistemically neutral and felicitous under any evidential context.

Various languages that are slated for possible inclusion in subsequent experiments (e.g. Georgian¹², Turkish¹³, and Bulgarian¹⁴) are noted for conveying evidentially in their PLGs specifically. In fact, as noted in the typological literature (e.g. Aikhenvald 2004), evidentiality is frequently grammaticalized in many languages, and in some cases is particularly associated with PLGs, and thus this feature is included in the schema used in this project.

As evidentiality is not grammatically encoded in the English TAM system, the handful of instances of this feature in the pilot experimental dataset all either explicitly encode source of knowledge (such as “he saw that [predicate]”) or otherwise make source of knowledge explicit in the discourse context.

2.2.1.6 TAM Layering

Tense-Aspect-Modality Layering (TAML) refers to grams in which multiple TAM operators co-occur within a single clause, often combining to yield complex interpretive effects. In the context

¹²Widely cited, incl. Bolkvadze & Kiziria, 2023; Kurtsikidze, 2006; Makharoblidze, 2012

¹³Sener (2011), Rett, Hyams, & Winans (2013)

¹⁴Rett & Murray (2013)

of PLGs, this commonly includes combinations such as perfect + progressive (e.g. *has been running*), perfect + past (e.g. *he had eaten*), or perfect + modal (e.g. *he might/should/could have arrived*). The feature schema used in this project treats such combinations as a distinct structural property with potential semantic effects. The operationalized definition of this feature which is used in the pilot study is as follows:

A gram incorporates TAM layering if it combines two or more tense-aspect-modality components (e.g., perfect + past, perfect + progressive, perfect + modal).

While not a distinct semantic category in and of itself, such as resultativity or evidentiality, TAML often interacts with semantic categories typologically associated with PLGs. In some languages, TAML is grammaticalized and semantically restricted, as in Japanese *-te ita* (Clarke, 2013; Nishiyama & Koenig, 2010) and Georgian *Pluperfect/turmeobiti II* (Bolkvadze & Kiziria, 2023; Kurtsikidze, 2006; Makharoblidze, 2012). Typologically, TAML is constrained but non-trivially variable, and PLGs are frequently at the center of such variation (Bybee 1994; Bohnemeyer, 2014; Cinque, 1999; Iatridou 2003).

This feature also reflects a broader orientation within this project, which treats PLGs less as isolated semantic readings (e.g. experiential, resultative) and more as structurally complex tense grams. This perspective informs both the typological framing and methodological design of the study.

In the pilot experiment this feature was a major source of classifier prediction errors; this will be discussed in more depth in the Results & Discussion section below.

2.2.1.7 Negation/Polarity Effects

While not a canonical diagnostic of PLGs, there is cross-linguistic evidence that polarity can interact with TAM expression, especially in systems with multiple PLGs (e.g. Georgian), such that negative polarity may favor one form over the other due to such factors as scope, inference blocking, and evidential contrasts (Michaelis, 1994; Portner, 2003). The operationalized definition of this feature which is used in the pilot study is as follows:

A token is marked as expressing negation if the gram appears under the scope of a negator, lexically, morphologically, or syntactically. This surface-level annotation is intended to assess whether gramal choice varies as a function of polarity.

The initial motivation to include this feature in project derives from the frequently cited (Aronson, 1990; Harris, 1981; Hewitt, 1996; Kurtsikidze, 2006; Makharoblidze, 2012) property of the Georgian Synthetic Perfect (Series III *turmeobiti I*) such that it is preferred over both the Aorist and Periphrastic Perfect under scope of negation. Georgian is considered a typologically interesting case study for this project as a rare language which bears multiple PLGs in its TAM system. As such the feature was incorporated on an exploratory basis.

In annotating polarity, I adopted a surface-oriented approach, marking grams as negative if they appear under the scope of an overt negator. However, it is worth noting that many non-veridical contexts, such as interrogatives, conditionals, and counterfactuals, also emerged in the pilot dataset and may warrant further investigation. These environments are traditionally associated with polarity sensitivity (Giannakidou, 1998; Israel, 2011), and it is plausible that they interact with TAM selection in subtle but systematic ways across languages.

While these contexts are not annotated separately in the current pilot, they are noted in the metadata and may serve as the basis for expanded polarity-related analysis in future iterations of the study.

2.2.1.8 Aktionsart

In addition to PLG features, the pilot experiment also annotated Aktionsart values (lexical aspect classes) for all tokens. Aktionsart captures the inherent temporal properties of predicates, such as whether they denote static situations (states), ongoing processes (activities), bounded processes with endpoints (accomplishments), or instantaneous changes (achievements) (Vendler 1957; Filip 2012). For instance, *know the answer* exemplifies a state, *run in the park* an activity, *build a house* an accomplishment, and *reach the summit* an achievement.

Tracking Aktionsart is essential because numerous studies show that lexical aspect interacts with grammatical aspect in systematic ways, shaping how TAM grams are distributed and interpreted cross-linguistically (Filip 2012; Cortés-Rodriguez 2016). In the context of PLGs, Aktionsart distinctions often condition the distribution of perfect-like grams, for example, accomplishments and achievements are more compatible with resultative readings, while states and activities may resist such interpretations or require coercion (M & M 2023). By incorporating Aktionsart annotations, this project is able to probe whether model mispredictions cluster around these semantic distinctions, offering a more fine-grained window into the interaction between lexical aspect and TAM morphology.

2.2.1.9 Honorary Mention: Current Relevance

A final note is warranted on the often-cited but methodologically elusive notion of “current relevance.” Especially in the context of the English present perfect, current relevance has long been proposed as a defining semantic contribution linking past events to present discourse salience (e.g. Comrie, 1976; McCawley, 1971; Michaelis, 1994). However, despite its ubiquity in the descriptive literature, there is no widely accepted diagnostic for identifying current relevance in context, nor consensus on whether it is part of the truth-conditional content of the gram at all (Grønn & von Stechow, 2021; Kiparsky, 2002; Portner, 2003; Schaden, 2009).

Because this study relies on discrete, scorable features that can be operationalized for cross-linguistic comparison, current relevance is not included in the present feature schema. Nonetheless, its persistence in the literature, and its relevance (current or otherwise) to how speakers and researchers conceptualize PLGs, warrants its mention here as a direction for further qualitative inquiry. Feedback and suggestions on this topic are very welcome.

2.2.2 Gram Identification

In this section I describe the grams observed in each of the languages of the study, how they were annotated, and the motivations behind their respective categorizations. In the context of this project, a gram refers to any morphosyntactic form or multimorphemic pattern that consistently conveys a temporal or aspectual meaning relevant to PLG classification. This includes overt TAM particles, light verb compounds, and other recurring predicate-internal forms. Many of the forms discussed here are technically composite grams, in that they combine multiple elements (e.g., past + perfect, modal + perfect, or *yǐjǐng* + *le*), in which each element itself is in fact a gram. For ease of exposition, I use ‘gram’ as a cover term in this section and throughout, but readers should bear in mind that a number of these grams are morphosyntactically composite.

While full translational equivalence is not assumed, aligned verses are treated as approximate functional equivalents for the purposes of gram comparison and reverse seeding. Recognizing the limitations of this assumption is especially important when evaluating cross-linguistic feature alignment.

2.2.2.1 English

As the anchor language used in this experiment, every token which was extracted from the corpus for evaluation was expressed in English as a perfect, i.e. auxiliary *have* + participle. These perfects were then divided by the specific TAM profile in which it was encoded, yielding six grams: [present] perfect, past perfect, *would* perfect, *could* perfect, *should* perfect, and *might* perfect. By far the most numerous were the perfect and past perfect grams; these were identified by the morphological tense expressed on the auxiliary, *have/has* for perfect and *had* for past perfects. No examples of progressive or other overt aspectual components were evident in the data (aside from the aspectual property conveyed by the perfect itself), though should such examples arise in subsequent experimental data, they will be annotated as such (creating a new gram designation for each, such as “present perfect progressive”), on the premise that it is easier to reduce distinctions later as needed than it is to re-annotate an entire dataset for distinctions that were not distinguished in the first pass.

Building on the same logic, each of the modal perfects was annotated distinctly, yielding a four-way distinction between *would*, *could*, *should*, and *might* perfects. The underlying motivation for distinguishing between these grams is that while each can be described as simply a “modal perfect”, it is possible that differences in cross-linguistic TAM systems may give rise to unforeseen interactions between modality and PLG selection. Based on the results of the pilot study, reflecting, as it does, a relatively small sample size of 300 total tokens (150 used for training and 150 for testing), it is unclear whether such fine-grained distinctions are in fact warranted in the grand scheme. As such, they will continue to be distinguished from one another until such time as experimental results either more robustly motivate their inclusion or indicate that they introduce more “noise” into the data than truly informative contrasts.

The list of English grams and the frequency of their occurrence in the 300-token data set is given in Figure 1.

Gram Name	Annotation	# Occurrences	% of tokens
Perfect	perf	215	72%
Past Perfect	pastperf	65	22%
<i>would</i> Perfect	wouldperf	13	4%
<i>should</i> Perfect	shouldperf	3	1%
<i>could</i> Perfect	couldperf	3	1%
<i>might</i> Perfect	mightperf	1	<1%
Progressive Perfect	progperf	1	<1%

Figure 1: List of English Grams

2.2.2.2 German

The German annotations recognized eight grams, each being associated with a specific morphological profile¹⁵: *Präsens*, *Präteritum*, *Perfekt*, *Plusquamperfekt*, *Konjunktiv I Perfekt*, *Konjunktiv*

¹⁵All grams are Indikativ (indicative) unless otherwise noted.

tiv II Präteritum, *Konjunktiv II Perfekt*, and *Konjunktiv II Plusquamperfekt*. grams in the data were identified using a combination of grammars and digital parsing tools (including Google Translate and the Reverso.net German conjugator). In a small number of cases, the corpus contained ‘orphan’ participles without an overt auxiliary. In these cases, the gram was coded as a *Perfekt*, with the tense component recovered from context. This decision was noted in the metadata. Because each gram is associated with a distinct semantic profile, they were kept distinct, though as with the English modal perfects, ongoing experimental results may indicate that some of these distinctions can (or should) be collapsed.

The list of German grams and the frequency of their occurrence in the 300-token data set is given in Figure 2.

Gram Name	Annotation	# Occurrences	% of tokens
Perfekt	perf	185	62%
Plusquamperfekt	pastperf	37	12%
Konj. II Plusquamperfekt	k2_pastperf	16	5%
Präsens	pres	12	4%
Konjunktiv II Präteritum	k2_past	10	3%
Präteritum	past	10	3%
Konjunktiv I Perfekt	k1_perf	5	2%
Konjunktiv II Perfekt	k2_perf	1	<1%

Figure 2: List of English Grams

2.2.2.3 Mandarin

Categorizing the Mandarin data proved to be somewhat challenging, as the TAM system it utilizes is quite distinct from what is found in the (vast majority of the) Indo-European languages on which much of the work on PLGs has been conducted, including those of strongly isolating languages such as English. Mandarin, rather than overtly marking tense in the conventional sense of definite time reference (i.e. past, present, future, non-past, etc.), it instead uses a variety of strategies for conveying the temporal properties of predicates.

One strategy that occurred quite frequently in the experimental dataset is what is annotated “nullmarked”¹⁶, i.e. a bare verb with no temporal situation or contour conveyed, either through morphological marking, aspectual particle, or essentially any other pronounceable element. Approximately 15% of tokens in the data utilized this strategy in correspondence with the English perfect. In the case of Mandarin, as well as other languages with similar TAM systems that utilize null TAM marking, a compelling puzzle arises: what is the *unmarked* interpretation of a predicate, with respect to the temporal properties? This facet of the linguistic system will be considered carefully in ongoing study.

¹⁶Note that “nullmarked” is distinct from “unmarked”; while a nullmarked form is overwhelmingly likely to be unmarked, a form may be unmarked but still overtly morphologically marked (i.e. and there for not) nullmarked, meaning that the relationship between these two concepts is a uni-directional (rather than bi-directional) entailment.

Another common strategy is the use of grammatical particles which are often referred to as “aspectual” in nature; the two most commonly associated with perfect-like meanings are *le* (often described as “perfective” or “completive”) and *guò* (sometimes referred to as “experiential”), both of which occur post-verbally. In addition to these, the light verb *yǒu* ‘have’ and its negative polarity counterpart *méiyǒu* are seen with some frequency in correspondence with the English perfect.

In addition to these, the gram *suǒ...de*, where the two elements more or less bookend the predicate, also appeared frequently in correspondence with English PLGs. Although not traditionally analyzed as a TAM marker, *suǒ...de* often functions as a passive or relativizing structure, which introduces a consistent and distinctive syntactic environment that may interact with PLG diagnostics, such as by foregrounding a delimited event without specifying resultant state. Its interaction with PLG-like interpretations remains an open question. The gram’s distribution and relevance will be further evaluated in future annotation phases, particularly with input from L1 Mandarin speakers, who are better positioned to assess whether it functions as a true perfect-like gram or simply as a frequent structural correlate. Examples of this gram are given in (16).

(16) Examples of Mandarin *suǒ...de*

- a. suǒ chī de
place eat of
‘that which is/was eaten’
- b. suǒ zhīdào de
place know of
‘that which is/was known’

While not strictly speaking a TAM marker, the gram does appear somewhat frequently in correspondence with the English perfect.

Mandarin also makes extensive use of adverbials to convey the temporal situation and contours of predicates. Many of these convey iative meanings, similar to English *already*, but come in a wide variety of subtle distinctions, including *yǐjīng* (neutral, “default” already-like meaning), *zǎoyǐ* (more emphatic and formal, with a remote past connotation), and *qǐlái* (less already, more directional/resultative with an inchoative dimension as well).

Further complicating the playing field, many of these components occur in tandem, such as *méiyǒu...guò*, *yǒu...le*, and [adverb] *le*. In the pilot annotation phase, each of these combinations of grams was annotated as a complex, single gram. The total number of distinctly annotated grams for Mandarin was 18; this diversity of in many cases relatively low-frequency grams no doubt added to the classifiers’ struggles in accurately predicting gram occurrence. Moving forward, this list of grams will be decomposed so as to better investigate each component independently.

It should be noted that roughly 8% of the Mandarin tokens in the experimental data were marked as “N/A”, as like with the “nullmarked” tokens they lacked overt TAM marking. These two groups were distinguished (somewhat informally, though as rigorously as possible given the limitations noted in the introduction to this methodology section) by whether some element of temporal/aspectual specification or contrast could be recovered from context. It remains unclear whether all nullmarked tokens genuinely lack aspectual specification, or whether some reflect covert imperfective interpretations. Distinguishing these cases is a key objective for future annotation refinement with L1 expertise.

Last but not least, the Mandarin TAM system employs a non-canonical strategy for aspectual specification whereby a “lexical aspect marker” (my term) is used to convey a specific aspectual

meaning. To clarify, these are not markers of lexical aspect, but rather lexical markers of aspect. In essence, how this system works is that a semantically contentful, usually monosyllabic root verb is joined with something like a light verb, often with directional/motion semantics, which thus “lexically” engenders a specific aspectual nuance. In the pilot experiment, all such examples, both those otherwise overtly marked and those not were omitted from consideration based on the relatively limited expertise of the annotator with respect to effectively evaluating the nuances conveyed. Again, this will be addressed going forward through incorporation of the expertise of L1 speakers.

The list of Mandarin grams and the frequency of their occurrence in the 300-token data set is given in Figure 3.

Gram Name	Annotation	# Occurrences	% of tokens
<i>le</i>	le	89	30%
<i>nullmarked</i>	nullmarked	40	13%
<i>suo...de</i>	suo_de	26	9%
<i>guo</i>	guo	12	4%
<i>yijing...le</i>	yijing_le	13	4%
<i>meiyoun...guo</i>	meiyoun_guo	13	4%
<i>meiyoun</i>	meiyoun	8	3%
<i>yijing</i>	yijing	8	3%
<i>zaoyi...le</i>	zaoyi_le	2	1%
<i>you...le</i>	you_le	2	1%
<i>qilai</i>	qilai	2	1%
<i>yi</i>	yi	2	1%
<i>yi...le</i>	yi_le	1	<1%
<i>yijing...you</i>	yijing_you	1	<1%
<i>zaoyi</i>	zaoyi	1	<1%
<i>zaojiu</i>	zaojiu	1	<1%

Figure 3: List of English Grams

2.2.3 Running the Experiments

Once all of the data had been annotated, they were fed into two classifier models for experimentation: Random Forests (“RFC”) and MLPClassifier (“MLP”). Random Forests operates on a decision tree model, while MLPClassifier (Multi-Layer Perceptron) is a neural network; using the two classifiers in parallel provides distinct perspectives on model performance, enabling comparison across model types and offering greater context for evaluating their salience and framing. In other words, a separate classifier was trained for each language, so that performance could be evaluated independently across language-classifier pairs.

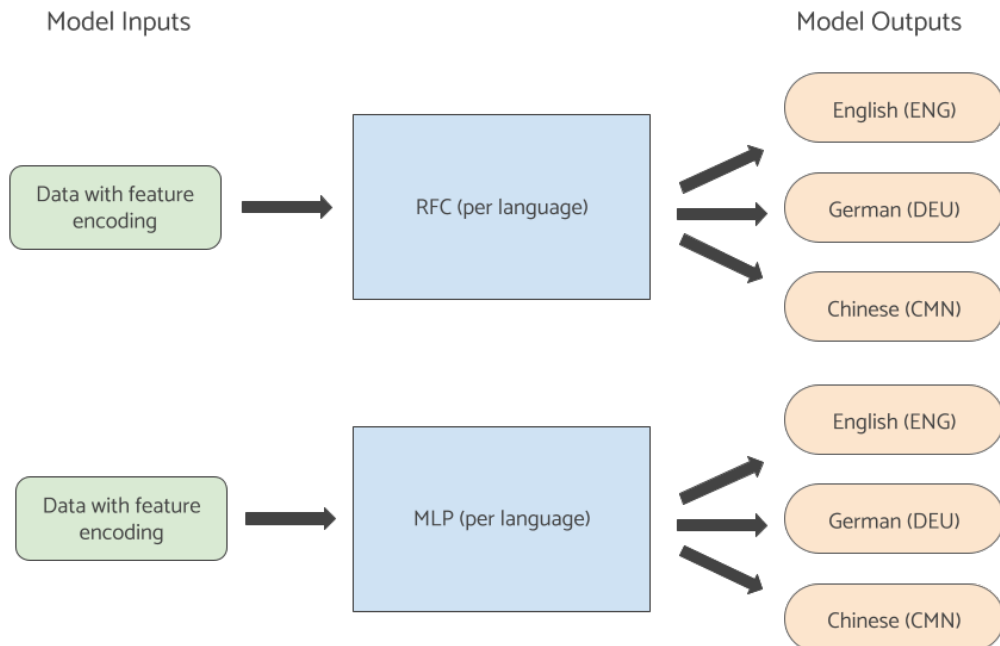


Figure 4: Workflow for classifier training and prediction. Feature-encoded data were used to train a separate Random Forest (RFC) and Multi-Layer Perceptron (MLP) classifier for each language (ENG, DEU, CMN), with outputs reported independently by language/classifier pair.

Both experiments used the same strategy for loading in the data: the 300 tokens were split into a 150-token training set and a 150-token test set. Although this pilot split was random rather than stratified, subsequent experiments will explicitly stratify by feature to ensure proportional distribution. The classifiers then used the 150-token training set to “learn” the features and correlation patterns with occurrence of grams, and based on that training attempted to predict which grams would be used in test set tokens. The results of the predictions were then exported for use in error analysis. Note that both the German and Mandarin datasets included tokens coded as “not applicable”, indicating that the token for that language was not usable for some reason; in subsequent experiments this binary applicable/ not-applicable distinction will be further decomposed into at least two subtypes of N/A: tokens in which a given verb/predicate is simply absent in that language’s text (“true” N/A), and tokens in which it is unclear whether the corresponding token “truly” encodes a PLG.

2.3 Results & Discussion

2.3.1 Overview

Before diving into language-specific results, one generalization can be made regarding both the models’ performance and also the methodology that produced the model inputs. The overall accuracy rates¹⁷ by language were, perhaps unsurprisingly, quite different. The English datasets both contained all 150 tokens, and performed with accuracy rates¹⁸ of 92.7% for RFC and 91.3%

¹⁷Accuracy is reported as the proportion of correctly predicted labels across all test instances.

¹⁸Here, ‘accuracy’ simply refers to the percentage of predictions that were correct.

for MLP. German accuracy was lower than that of English, with accuracy rates of 75.8% for RFC and 76.4%. Both models achieved significantly lower accuracy on the Mandarin data, with accuracy rates of 54.8% for RFC and 40.4% for MLP. There are a variety of factors which may be affecting this discrepancy between languages; here I highlight two specifically: anchor language effects and feature system effects.

2.3.1.1 Anchor Language Effects

Because English was used as the anchor language, the dataset contains a fuller set of English Perfect tokens, while German and Mandarin tokens appear only where they align with an English Perfect. This means the German and Mandarin samples represent a more restricted subset of their PLG inventories. Such asymmetry could potentially affect classifier performance (for example, by giving the models a richer picture of English than of the other languages) but the extent of this effect is ultimately an empirical question.

In addition, feature values for all tokens were assigned based on the English context. As a result, each feature bundle reflects the semantic features expressed by the English Perfect in that context, rather than necessarily those encoded in the corresponding German or Mandarin gram. This creates the possibility that non-English grams may carry features not represented in the annotation (e.g. morphosyntactic or lexical properties absent in English), or vice versa, introducing systematic mismatches in cross-linguistic feature profiles.

Subsequent experiments will explore ways to mitigate anchor language effects at both the selection and annotation levels. These include incorporating reverse seeding across all languages, soliciting input from L1 speakers, and refining the feature inventory to better capture language-specific distinctions. While such measures cannot fully eliminate anchor-based biases, and may, in some cases, redistribute rather than remove them, they nonetheless represent crucial steps toward testing the cross-linguistic robustness of the proposed methodology across typologically diverse languages.

2.3.1.2 Feature System Effects

As mentioned in the previous section, with respect to the Mandarin TAM system, it generally goes without saying that each language’s TAM (in fact, entire grammatical system) operates on its own logic, including what features it chooses to contrast or mark overtly versus those it doesn’t¹⁹, the number of TAM grams in the system’s inventory, and more. Because the bulk of the research conducted on PLGs to date has focused on Indo-European, and especially Germanic languages (English alone represents a significant focus of PLG research historically), the feature set that was developed for the pilot experiment, reflecting the research system, tends to better reflect the types of features that those languages “care about” most. For example, if the English perfect is considered a “complex tense”²⁰, i.e. in the Reichenbach (1947) sense of a grammatical form which combines tense and aspect, or which divides its event and reference times, to what extent can we truly apply the logic developed for English to a language which, like Mandarin, has been called “tenseless” (Lin, 2006)?

Tackling this quandary is one of the central aims of this project, by way of developing a feature set which can be used effectively on cross-linguistic and typologically varied languages.

¹⁹Of course, this reflects an etic perspective (researcher-imposed categories) rather than an emic one (language-internal categories).

²⁰This is my own term, but I’m quite fond of it. I am unsure to what extent it is used elsewhere in the literature; I have seen it, but not a lot.

In the next section I present the results by language along with some discussion, followed by some global/cross-linguistic performance discussion.

2.3.2 Results & Interpretation by Language

This section presents the results of each language-classifier experiment. The primary metric is overall performance accuracy (how well did the classifier do globally, what is its success rate), which I will examine using several key methods, including specific error types, feature-error type associations, and several statistical evaluations (including χ^2 scores).

2.3.2.1 Performance Metrics

To evaluate classifier performance, a series of complementary metrics were used. Accuracy provides a broad measure of overall performance, while error type breakdowns illustrate the distribution of misclassifications across grams. Feature-based metrics (FER, WS) assess whether errors cluster around specific semantic features relative to baseline expectations, while χ^2 , p , and ϕ assess the reliability and strength of such associations. Finally, feature-error association tables combine these perspectives to highlight systematic links between particular error patterns and feature bundles. Note that where quantitative counts and proportions are given, these numbers refer to the test dataset only; training data is not included in these values.

The first measure provided for each language is the overall accuracy rate by classifier, calculated as the number of correct predictions divided by the total number of predictions. For example, an accuracy of 75% means that three out of four predictions were correct, while one out of four were erroneous. This provides a simple sanity check on model performance before turning to more detailed analyses of error patterns and feature effects.

The next set of metrics focus on specific error types, represented as *predicted* $\rightarrow$ *actual*²¹. For each type, both the count and its percentage of all errors are reported. For example, in the Mandarin RFC data, the error type *le* $\rightarrow$ *nullmarked* occurred eight times, meaning that nullmarked was misclassified as *le* in eight tokens, or 15% of all errors in that experiment. Unlike overall accuracy, which gives a single performance score, error-type metrics highlight where the model struggles most, revealing systematic confusions between particular grams.

Following the error-type metrics, feature performance metrics are reported using a custom measure developed for this experiment called the Weighted Score (WS). The first step is to calculate the baseline error rate (BER), defined as the total number of classification errors divided by the total number of tokens in the test set. Importantly, this denominator includes all tokens in the dataset, not only those carrying a given feature. BER therefore reflects the error rate we would expect if errors were evenly spread across all features, providing a neutral yardstick for comparison.

The next step is to calculate the feature-specific error rate (FER), defined as the number of errors involving a given feature divided by the total number of tokens that carry that feature. FER shows how often the model got tokens with that feature wrong, giving a direct measure of the feature’s "difficulty". The last component we will need is the feature’s prevalence (PREV), the prevalence of the feature in the dataset, is the proportion of all tokens in the test set that carry that feature; this value is what we will use to weight the final score.

Once the BER, FER, and PREV are determined, it is possible to assign a Weighted Score (WS) to each feature. To find the WS, we first subtract the BER from FER, which compares "how hard this feature is" relative to model’s average performance (e.g. if the FER is higher than than

²¹This is often referred to by the term "gold"; here I favor "actual" as it is more intuitively accessible to a non-specialist audience.

the BER, then the result will be higher, indicating that the model produced more errors when the feature was present than it would have if all errors were distributed evenly among features). This number is then multiplied by the feature’s PREV in order to scale that difference by how frequent the feature actually is in the dataset. This step corrects for the different occurrence rates of individual features, such as the difference between the Resultative feature, which occurred in 83 of the Mandarin RFC experiment’s test dataset, and the DTAM feature, which occurred one time in the same dataset. Finally, this value (the Base Weighted Score or BWS) is multiplied by 100 in order to scale for interpretability, giving the final WS. The calculations for each of these values are provided in (17) below.

- (17) a. Baseline Error Rate (BER): $\frac{\text{total errors}}{\text{total tokens}}$
b. Feature-Specific Error Rate (FER): $\frac{\text{errors that bear the feature}}{\text{total occurrences of feature in test set}}$
c. Prevalence (PREV): $\frac{\text{total occurrences of feature in test set}}{\text{total tokens in the test set}}$
d. Base Weighted Score (BWS): $(\text{FER} - \text{BER}) * \text{PREV}$
e. Final Weighted Score (WS): $\text{BWS} * 100$

Using these calculations, we can assess the extent to which each individual feature drove errors in the classification experiment by virtue not only of how often it was associated with erroneous predictions, but also by how frequently it occurred in the test dataset. In other words, the WS shows whether a given feature systematically drove errors in a given experiment. The actual numeric value of WS is interpreted as a deviation from BER, weighted by prevalence. In practice:

- (18) a. $\text{WS} = 0$: the feature did not deviate from BER, or hypothetical even distribution of features across errors, for that experiment.
b. $\text{WS} > 0$: the feature was “harder” for the classifier, i.e. it deviated such that its error rate was **higher** than BER.
c. $\text{WS} < 0$: the feature was “easier” for the classifier, i.e. it deviated such that its error rate was **lower** than BER.

The features with the most marginal scores (i.e. the highest and lowest WS values) were then subjected to a two-tailed χ^2 test²² to evaluate their statistical impact on model performance. This test was conducted on a 2x2 contingency table comparing tokens that bore a given feature with whether they were misclassified. For example, {f+, e+} represents the number of tokens that carried the feature and were misclassified, while {f-, e+} represents tokens that lacked the feature but were still misclassified. The χ^2 statistic, along with its associated p -value, indicates whether there is evidence of association between feature presence and model errors, and how confident we can be that the association is not due to chance; here $p\text{-value} \leq 0.05$ is considered statistically significant. Because χ^2 and p -values are influenced by sample size (N), effect size is also reported using the ϕ coefficient (ϕ), which measures the strength of association on a scale from 0 (no association) to 1 (perfect association). While ϕ is not entirely independent of N, it scales χ^2 by N, making it less inflated by large sample sizes than χ^2 alone, though still unstable if N is very small.” The formula used to calculate the ϕ coefficient is given in (19), along with its scoring ranges (20). Taken together, the χ^2 statistic, its associated p -value, and the ϕ coefficient indicate whether

²²This analysis incorporated the Yates correction, which is designed to correct for smaller sample sizes to avoid overstating significance values.

a feature is associated with errors, how confident we can be that this association is not due to chance, and how strong the association is.

$$(19) \quad \phi = \sqrt{\frac{\chi^2}{N}}$$

(20) ϕ coefficient ranges

- a. $\phi < 0.3$ = weak association
- b. $0.3 \leq \phi < 0.5$ = medium association
- c. $\phi \geq 0.5$ = strong association

These ranges provide an interpretive scale for ϕ , helping to classify the strength of association observed between features and errors.

Finally, the metrics above (accuracy, error types, feature metrics, and statistical measures) are synthesized to determine which features are associated with which error types, and the degree to which those associations are statistically significant. For each language-classifier pair, a table reports: the error types (predicted $\rightarrow$ actual), the associated feature bundles, and the percentage of errors of each type that share the listed feature bundle. For a more complete assessment of why certain errors occur, the frequency of occurrence of the feature (combinations) in the two gram combinations involved in the error pattern would have to be compared as well. Such comparison, to be worked out in a subsequent iteration, would allow us to assess if a given error reflects ‘odd’ tokens of the true category with respect to the feature value, or rather other explanations (like frequency) for cases where the feature is prevalent for both categories.

The metrics used in this study are summarized in (21) for ease of reference.

(21) Summary of statistical metrics

- a. **Accuracy** = correct predictions $\div$ total predictions (overall performance)
- b. **Error types** = predicted $\rightarrow$ actual, with counts and percentages of total errors (where models struggle most)
- c. **Weighted Score (WS)** = (FER - BER) * PREV * 100, comparing feature error rate against expected baseline, weighted by prevalence (shows which features are harder or easier)
- d. χ^2 , p , ϕ = statistical tests of association: χ^2 (whether association exists), p -value (confidence it’s not chance), ϕ (strength of association)

2.3.2.2 Abbreviations

In the results below, it will at times be useful to refer to features by abbreviations, in which cases the following conventions will be used.

- Experiential: Exp
- Resultative: Res
- Universal: Univ
- DTAM: (same)
- TAM Layering: TAML
- Negative Polarity: NegPol

- Activity: Act
- Accomplishment: Acc
- Achievement: Ach
- State: (same)
- Semelfactive: semel

2.3.2.3 Mandarin Results

The RFC experiment achieved an overall accuracy of 55% on the Mandarin data, reflecting 63 accurate predictions out of 115 total predictions. The MLP experiment achieved an overall accuracy of 50% on the Mandarin data, reflecting 61 accurate predictions out of 121 total predictions. The results point to a core asymmetry that will be unpacked below: both models appear to identify resultative PLGs more readily than experiential ones, raising questions about the interaction between surface cues and feature ambiguity under the current schema.

2.3.2.3.1 Major Error types

Predicted → Actual	Count	% of All Errors
le → suo...de	7	13%
le → nullmarked	8	15%
le → yijing...le	8	15%
nullmarked → le	7	13%
nullmarked → (varies)	6	12%
Totals:	36	69%

Figure 5: Major Mandarin RFC error types, alongside gram occurrence rates in test set

Predicted → Actual	Count	% of All Errors
le → suo_de	9	15%
le → nullmarked	8	13%
le → yijing_le	8	13%
suo_de → (varies)	14	23%
nullmarked → le	5	8%
nullmarked → (varies)	5	8%
Totals:	49	82%

Figure 6: Major Mandarin MLP error types, alongside gram occurrence rates in test set

Both classifiers show a tendency to overgeneralize on the usage of *le*, which accounted for 43% of all RFC errors and 41% of all MLP errors. Importantly, *le* itself represented only about 35% of the test sets, meaning the models predicted it at a rate slightly higher than its actual frequency. Strikingly, all 31 RFC *le* misclassifications and all 33 MLP *le* misclassifications occurred in resultative contexts, a pattern that will be examined in more detail in the Feature Metrics section below. Similarly, 25% of RFC errors and 17% of MLP errors were overpredictions of nullmarked grams (i.e. cases where the model predicted nullmarked incorrectly), despite nullmarked grams making up only 11% of the test set. Put differently, the models produced a high number of false positives for *le* and nullmarked, meaning these grams were predicted more often than expected given their actual prevalence in the test set. By contrast, the handling of *suo...de* showed a different pattern: although it appeared in just 7% of the dataset, MLP misclassified it in 23% of its errors, while RFC produced only three such errors (one misclassified as *guo* and two as nullmarked). This asymmetry suggests that the two classifiers may be responding to different cues in applying the feature schema, though other factors, such as stochastic variation in training or differences in how errors are distributed across categories, could also contribute. The 14 MLP errors of this type broke down as five *le*, three *guo*, three nullmarked, two *meiyou...guo*, and one *zaoyi* (an “already”-like iative with remote past connotations). Finally, the nullmarked → (varies) errors were widely distributed across gram actuals, showing no discernible pattern.

2.3.2.3.2 Feature Metrics

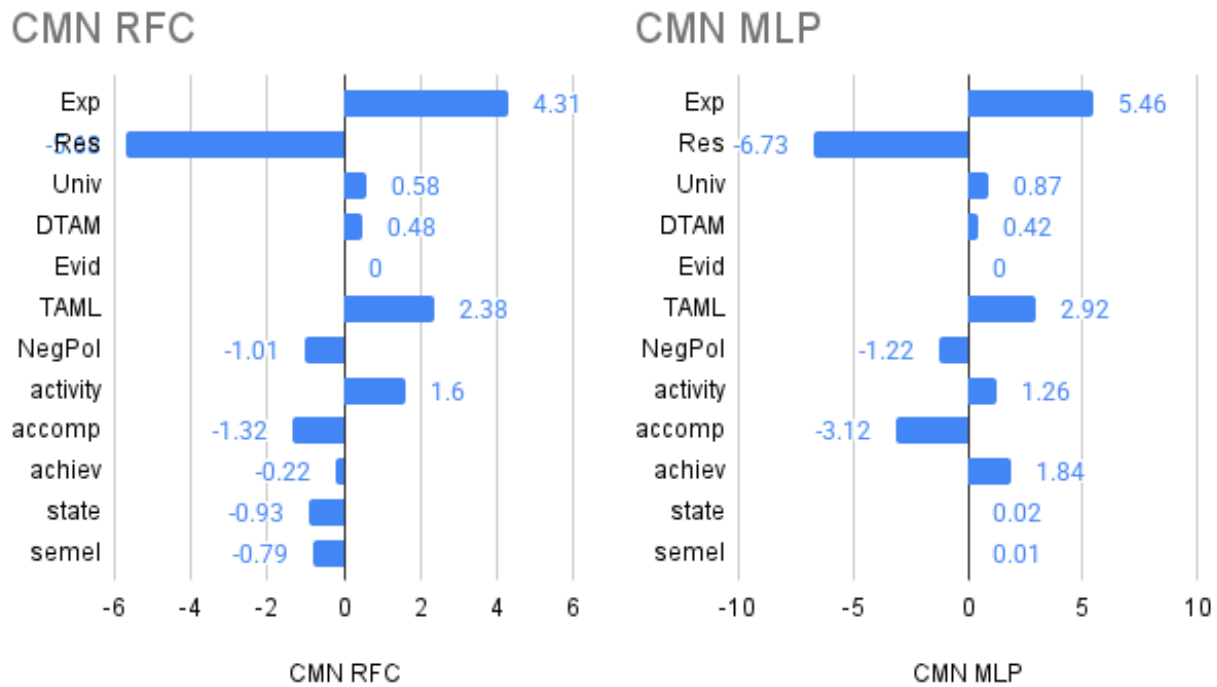


Figure 7: Mandarin Weighted Scores (WS) By Feature

	Random Forest (RFC)		MLPClassifier (MLP)	
	FER%	WS	FER%	WS
Experiential	70%	4.31	74%	5.46
Resultative	37%	-5.68	40%	-6.73
Universal	50%	0.58	58%	0.87
DTAM	100%	0.48	100%	0.42
Evidentiality	0%	0.00	0%	0.00
TAML	100%	2.38	100%	2.92
NegPol	29%	-1.01	20%	-1.22
Activity	71%	1.60	71%	1.26
Accomplishment	42%	-1.32	43%	-3.12
Achievement	45%	-0.22	54%	1.84
State	33%	-0.93	50%	0.02
Semelfactive	0%	-0.79	50%	0.01

Figure 8: Mandarin Feature Metrics for Random Forests and MLPClassifier

The Mandarin results reveal a strong asymmetry: while grams bearing the Resultative feature were among the easiest for both models, those bearing the Experiential feature were markedly more difficult, a pattern not observed in the other languages. Neither German nor English showed such a clear feature-driven distinction between PLG types. Other features in the Mandarin experiment were close to 0, meaning their error rates showed no evidence of being systematically easier or harder than the average across features see Figure 8. RFC results were slightly more clustered, while MLP distributed difficulty more broadly across features. Effect sizes are weak, so interpretations remain tentative.

The tables above show overall feature performance by classifier: recall that lower WS values indicate stronger performance on that feature. Figure 9 presents the statistical measures for the features deemed most significant by χ^2 analysis, showing the degree to which feature performance rates impacted overall accuracy.

	p -value	ϕ -value
Resultative, RFC	0.0117	0.255
Resultative, MLP	0.0024	0.295
Experiential, RFC	0.0276	0.229
Experiential, MLP	0.0076	0.263

Figure 9: Mandarin Statistical Measures

For both models, the Resultative and Experiential features were deemed statistically significant, though ϕ -values indicate weak association between errors and features, with the exception of the Resultative feature in the MLP experiment, which at $\phi = 0.294$ is just below the conventional threshold (0.3) for a medium-strength association.

Taken together, the statistical results suggest only weak associations overall, but several error types nonetheless align systematically with particular feature values. The tables and rough interpretations below highlight these associations, offering a bridge between the quantitative metrics and the interpretive discussion in the next section. →

Error Type	Feature Association(s)	% Error Match
<i>le</i> → <i>suo...de</i>	7/7 +Res, 5/7 + Acc	71%
<i>le</i> → <i>nullmarked</i>	8/8 +Res, 6/8 +Ach	75%
<i>le</i> → <i>yijing...le</i>	7/8 +Res	88%
<i>nullmarked</i> → <i>le</i>	7/7 -Res	100%
<i>nullmarked</i> → (<i>varies</i>)	6/6 -Res	100%

Figure 10: Mandarin RFC Error Type-Feature Associations

Error Type	Feature Association(s)	% Error Match
<i>le</i> → <i>suo...de</i>	8/8 +Res, 6/8 + Acc	75%
<i>le</i> → <i>nullmarked</i>	8/8 +Res, 6/8 + Ach	75%
<i>le</i> → <i>yijing...le</i>	8/9 +Res	89%
<i>nullmarked</i> → <i>le</i>	5/5 -Res	100%
<i>nullmarked</i> → (<i>varies</i>)	5/5 -Res	100%
<i>suo...de</i> → (<i>varies</i>)	14/14 +Exp	100%

Figure 11: Mandarin MLP Error Type-Feature Associations

Rough interpretations:

- *le* → *suo...de* and *le* → *nullmarked* errors suggest over-association of *le* with Resultative feature and possible Aktionsart interaction
- *le* → *yijing...le* errors are likely not “true” errors (adverb ignored?)
- *nullmarked* → (*varies*) errors support +/-Resultative split

- *suo...de* $\rightarrow$ (varies) errors account for 56% of all MLP experiential errors; unclear why it occurs only in the MLP experiment

2.3.2.3.3 Discussion

Several important insights can be gleaned from the Mandarin experiments. First, the models appear to interpret the TAM system as organized primarily around a +/-Resultative distinction. This is reflected in the feature metrics: resultatives are among the easiest features to predict (WS = -5.68 for RFC; -6.73 for MLP), while experientials are consistently difficult (WS = 4.31 for RFC; 5.46 for MLP, where positive WS indicates above-baseline error). The majority of high-frequency errors involve *le* grams, including *le* $\rightarrow$ *suo...de*, *le* $\rightarrow$ nullmarked, and *le* $\rightarrow$ *yijing...le*, each accounting for 13-15% of total errors in both models. These confusion patterns disproportionately involve resultative features (e.g. +Res, +Acc), suggesting that *le* is often misaligned with strongly resultative contexts. Associations between error types and Aktionsart (notably accomplishments and achievements) hint at a semantic basis for these patterns, though statistical analysis indicates weak effect sizes (χ^2 results showed no Aktionsart feature-error association), and such claims remain tentative.

It should be noted that the *le* $\rightarrow$ *yijing...le* errors may not count as “true” errors: the models consistently predict *le* while omitting the co-occurring adverb. This suggests that the models treat the adverb as non-consequential, a hypothesis to be tested by decomposing the morphemic components, as well as by using larger datasets, a refined feature schema, and insights from L1 speaker data.

The nullmarked $\rightarrow$ (varies) error is noteworthy because every instance of this error type was negative for the Resultative feature (errors were evenly split between experientials and universals in both experiments). In the RFC dataset, 83 of 115 (72%) tokens were +Resultative, and in the MLP dataset 85 of 121 (70%). These distributions are nearly complementary, with exceptions limited to 15/300 ambiguous Experiential/Resultative tokens overall (3/115 in the Mandarin experiment specifically). This supports the hypothesis that the models interpret Mandarin PLGs as operating on a +/-Resultative distinction, though whether this reflects the actual TAM system will be investigated further in Project 2.

The *suo...de* $\rightarrow$ (varies) error appears only in the MLP experiment. Every token in this set is experiential, representing 56% of all prediction errors with the Experiential feature. Combined with a low *p*-value for the Experiential feature in the Mandarin MLP dataset, this suggests that experientiality is highly salient to the model, though the low ϕ -value warrants caution in interpretation.

The *le* $\rightarrow$ nullmarked and nullmarked $\rightarrow$ *le* errors reveal a striking asymmetry in feature associations. All *le* $\rightarrow$ nullmarked errors (8/8 in both models) involve +Resultative contexts, with a majority also associated with achievements. This suggests that the models sometimes fail to recognize *le* as marking resultative aspect, instead defaulting to a null form despite strong semantic cues. In contrast, all nullmarked $\rightarrow$ *le* errors (7/7 in RFC; 5/5 in MLP) occur in -Resultative contexts, indicating that *le* is also occasionally overpredicted where no resultative interpretation is present. A similar pattern holds for nullmarked $\rightarrow$ (varies) errors in the MLP dataset, which also show uniformly -Resultative feature values. Together, these patterns suggest that *le* functions as a kind of “gravitational center” in the TAM space: both under- and over-predicted in ways that track closely with the Resultative feature, but not always with sufficient sensitivity to contextual cues. Further feature refinement and error-type expansion may help clarify this interpretive instability.

From a linguistic perspective, one plausible takeaway these results is that the feature schema in its current state struggles to capture cross-linguistic nuances in the interpretations of Mandarin

PLGs; after all, much existing PLG research focuses on Indo-European (especially Germanic) languages, leaving non-Indo-European systems underexplored by comparison. Conversely, this may reflect a limitation of the annotation strategy used in the pilot experiment generally: many gram components were conflated into complex forms (e.g., [adverb] + *le*, and *you* + *le*). The next phase of testing will split these complexes into their component parts for individual evaluation, allowing the schema to be refined in tandem with clearer gramal units.

2.3.2.4 German

The RFC experiment achieved an overall accuracy of 76% on the German data, reflecting 113 accurate predictions out of 149 total predictions. The MLP experiment also achieved an overall accuracy of 76% on the German data, reflecting 113 accurate predictions out of 148 total predictions. The German results reveal a characteristic profile: both models are destabilized by TAM layering (pastperf), show patterned misinterpretations of universals depending on predicate class, and yet handle resultatives with notable ease, suggesting a mix of feature salience and schema-level limitations.

2.3.2.4.1 Major Error types

Predicted → Actual	Count	% of All Errors
pastperf → k2_pastperf	7	19%
pastperf → past	4	11%
pastperf → perf	4	11%
perf → pres	4	11%
perf → (varies)	7	19%
pres → (varies)	4	11%
Totals:	26	84%

Figure 12: Major German RFC error types, alongside gram occurrence rates in test set

Predicted → Actual	Count	% of All Errors
pastperf → k2_pastperf	6	17%
pastperf → past	4	11%
pastperf → perf	4	11%
perf → pres	4	11%
perf → (varies)	7	20%
pres → (varies)	4	11%
	29	83%

Figure 13: Major German MLP error types, alongside gram occurrence rates in test set

The two models performed nearly identically with respect to their major error types. One particularly noteworthy pattern is the tendency of both models to overgeneralize the use of *pastperf*, which accounts for only 8% of the German test sets but drives roughly half of all errors in both experiments. This indicates that the models are disproportionately sensitive to *pastperf*, overpredicting it well beyond its actual frequency in the data.²³ By contrast, *perf* accounts for the majority of the German dataset (63% of tokens), so its high error rate (31% for both models) is less surprising: the models are exposed to *perf* much more often, and small proportional misclassifications add up quickly. Taken together, these patterns highlight how error proportions can only be properly understood relative to gram base rates: *pastperf* is rare yet disproportionately error-prone, while *perf* is frequent and errors reflect this prevalence.

²³These percentages include overpredictions of *pastperf* which are not represented in these tables.

2.3.2.4.2 Feature Metrics

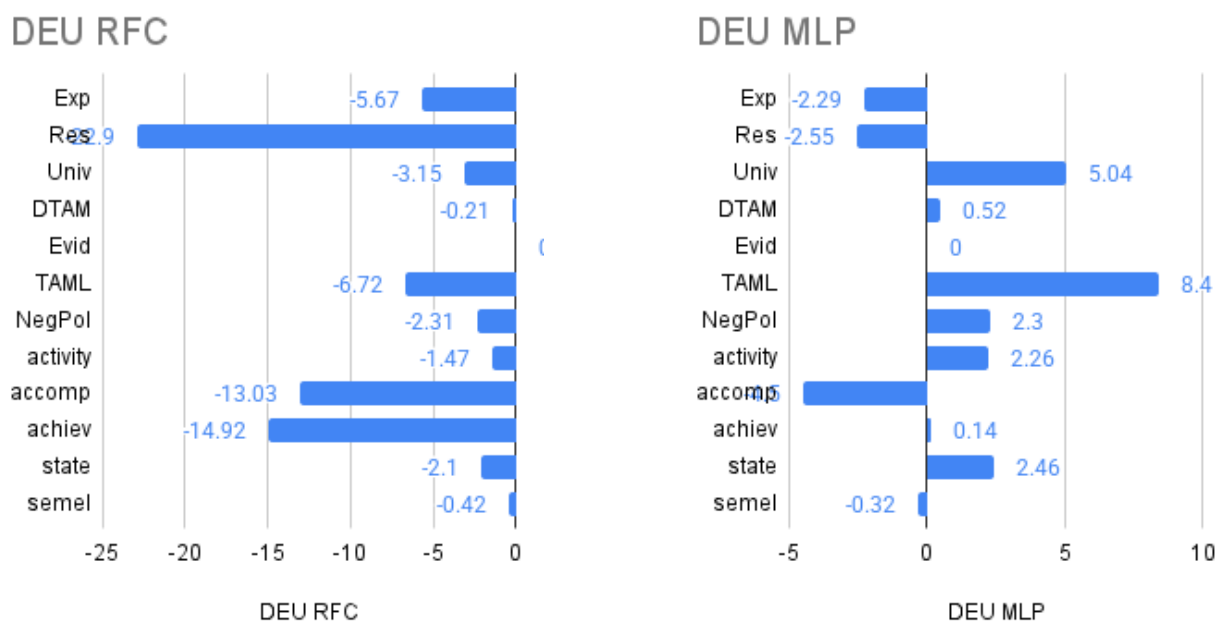


Figure 14: German Weighted Scores (WS) By Feature

	Random Forest (RFC)		MLPClassifier (MLP)	
	FER%	WS	FER%	WS
Experiential	11%	-5.67	11%	-2.29
Resultative	21%	-22.90	20%	-2.55
Universal	73%	-3.15	73%	5.04
DTAM	100%	-0.21	100%	0.52
Evidentiality	0%	0.00	0%	0.00
TAML	64%	-6.72	63%	8.40
NegPol	55%	-2.31	55%	2.30
Activity	75%	-1.47	71%	2.26
Accomplishment	13%	-13.03	13%	-4.50
Achievement	24%	-14.92	24%	0.14
State	60%	-2.10	60%	2.46
Semelfactive	0%	-0.42	0%	-0.32

Figure 15: German Feature Metrics for Random Forests and MLPClassifier

Perhaps most striking in the feature metrics presented here is the difference in the ranges of the models' WS. While MLP performed similarly on German as on Mandarin, both in terms

of range size and bookend values, RFC performed significantly better than baseline (0.24 for both models) on all features. Much of this variance is driven by the Resultative feature, which RFC handled very well, and to a lesser extent the Accomplishment and Achievement features.

The WS range for the German RFC experiment was -22.90 to 0.00 (22.90), indicating that all features scored at or below the model’s overall baseline error rate (0.24 for this experiment). In other words, the model performed at least as well on each individual feature as it did on average across the dataset. However, this apparent uniformity masks a sharp skew: some features were considerably easier than average, while others were only slightly below. Most notably, the Resultative feature scored a WS of -22.90, showing that it was by far the most predictable category in the experiment. By contrast, features like DTAM (-0.21) and Activity (-1.47) were only marginally easier than average, while others such as Accomplishment (-13.03) and Achievement (-14.92) fell between these extremes. This shows that the RFC model concentrates its predictive strength on a small set of highly salient features, while performance on other features remains closer to baseline.

By contrast, the WS range for the German MLP experiment was -4.50 to 8.40 (12.90), indicating broader differentiation across features against a baseline of 0.24 for this experiment. In this case, some features performed worse than the baseline error rate while others performed better, yielding what appears to be a more balanced profile than RFC. However, this “balance” does not necessarily reflect better learning. Rather, it suggests that MLP distributes its attention more widely across the feature space, sometimes overextending into features that the model does not exploit productively. RFC, in contrast, concentrates more narrowly on a few highly predictive cues (particularly resultatives), which creates asymmetry but may also reflect greater efficiency in how the feature schema is being used. That both classifiers achieved the same accuracy despite these differences points to important model-specific pathways: MLP achieves accuracy through broad but diffuse feature use, while RFC achieves it through sharper but narrower reliance on a few dominant features.

The models’ differential performance almost certainly reflects sensitivity to linguistic cues not captured in the experimental feature annotations or metadata, including finer-grained lexical, morphological, or discourse-level patterns. Future work will incorporate a more nuanced assessment of these factors, beyond what was feasible within the compressed timeline of the pilot experiment.

Figure 16 shows the statistical measures for the features deemed most significant by χ^2 analysis, informing the degree to which the feature performance rate impacted overall accuracy.

	p -value	ϕ -value
TAML RFC	0.000	0.505
TAML, MLP	0.000	0.480
Universal, RFC	0.000	0.394
Universal, MLP	0.000	0.393
Activity, RFC	0.002	0.289
Activity, MLP	0.010	0.251
State, RFC	0.015	0.231
State, MLP	0.016	0.230
Accomplishment, RFC	0.017	0.211
Accomplishment, MLP	0.017	0.211
Negative Polarity, RFC	0.031	0.207
Negative Polarity, MLP	0.033	0.206

Figure 16: German Statistical Measures

For both models, the TAML and Universal features were deemed very highly significant, with TAML ϕ -values at or near the threshold for strong association strength, and Universal ϕ -values well within the medium association strength range. While the remaining features were all deemed statistically significant, the association strength for all is weak.

To synthesize the German feature results, Figures 17 and 18 align major error types with their feature associations and interpretations. This summary illustrates how errors cluster around specific TAM bundles (especially those involving *pastperf*), and where the models appear to blur functional boundaries.

Error Type	Feature Association(s)	% Error Match
<i>pastperf</i> $\rightarrow$ <i>k2.pastperf</i>	7/7 +Res, 7/7 +TAML, 5/7 +Ach	71%
<i>pastperf</i> $\rightarrow$ <i>perf</i>	4/4 +Res, 4/4 +TAML, 4/4 +Ach	100%
<i>pastperf</i> $\rightarrow$ <i>past</i>	4/4 +Res, 4/4 + TAML, 3/4 +Acc	75%
<i>perf</i> $\rightarrow$ <i>pres</i>	4/4 +Univ, 4/4 +Act	100%
<i>pres</i> $\rightarrow$ (<i>varies</i>)	4/4 +Univ, 4/4 + State	100%
<i>perf</i> $\rightarrow$ (<i>varies</i>)	7/7 None	100%

Figure 17: German RFC Error Type-Feature Associations

Error Type	Feature Association(s)	% Error Match
<i>pastperf</i> → <i>k2_pastperf</i>	6/6 +Res, 6/6 +TAML, 5/6 +Ach	83%
<i>pastperf</i> → <i>perf</i>	4/4 +Res, 4/4 +TAML, 4/4 +Ach	100%
<i>pastperf</i> → <i>past</i>	4/4 +Res, 4/4 + TAML, 3/4 +Acc	75%
<i>perf</i> → <i>pres</i>	4/4 +Univ, 4/4 +Act	100%
<i>pres</i> → (varies)	4/4 +Univ, 4/4 + State	100%
<i>perf</i> → (varies)	None 7/7	100%

Figure 18: German MLP Error Type-Feature Associations

Rough interpretations:

- *pastperf* → *k2_pastperf* errors suggest that mood distinction (Indikativ versus Konjunktiv II) is opaque and/or overgeneralization of TAML feature
- *pastperf* → *past* and *pastperf* → *perf* errors track Aktionsart; also support overgeneralization of TAML feature
- *perf* → *pres* and *pres* → (varies) errors suggest (mis)analysis of Aktionsart in +Univ contexts
- *perf* → (varies) errors are most likely due to frequency-drive overgeneralization of the Perfekt

2.3.2.4.3 Discussion

The German results highlight three core insights. First, both models struggled with TAM layering, with the *pastperf* gram heavily overpredicted in ways that correlate directly with the TAML feature. Second, the models treat universals as a salient category, but Aktionsart interacts with them in misleading ways: activities are pulled toward present and states toward perfect. This suggests that while the models recognize universals as meaningful, they misinterpret how predicate class interacts with TAM morphology in German. Third, both models performed notably well on resultatives, though statistical analyses did not confirm this feature as significant. Together, these findings suggest that while the models interpret core features reliably, they face difficulty when those features interact in layered or aspectually nuanced ways.

The overprediction of *pastperf* is especially striking: every instance correlated directly with the TAML feature, which was both statistically significant and bore a strong association. This suggests that TAM layering is systematically difficult for the models to capture. One likely reason is the coarseness of the TAML feature itself, which conflates multiple distinct phenomena, in this dataset, past and modal perfects, essentially reading every instance of complex morphological expression (e.g. past + perfect, modality + perfect, etc.) as identical. This points to the need for decomposing

TAML into more fine-grained categories such as “tense-shifted” and “modal,” or even subtypes of modality. Another likely factor is, as with the Mandarin data, multiple layers of morphological expression are conflated in the annotation scheme, such as verbal mood (Indikativ, Konjunktiv I & II), which can in future experiments be further decomposed to better evaluate the independent performance of each component.

The universal errors provide another window into model interpretation. Both models consistently confused *perf* and *pres* in the presence of the Universal feature, but in a way that split along Aktionsart lines: universal activities skewed toward *pres*, while universal states skewed toward *perf*. This indicates that the models’ handling of universals is not random but conditioned by predicate class, with eventive predicates more stably tied to present and stative predicates more often pushed toward perfect. Why this should be is a matter that is currently under consideration.

Resultatives, by contrast, were handled especially well (particularly by RFC), despite statistical analyses not assigning the feature significance. This mismatch suggests that distributional cues, such as frequency and clear morphosyntactic signals, may be sufficient for the models to detect and reliably exploit resultativity in prediction, even if statistical measures do not flag significance.

Finally, the *perf* → (varies) errors stand out for their lack of systematic feature associations. These cases likely reflect an overgeneralization to the frequent *perf* gram in contexts where feature profiles were uninformative. In other words, *perf* functions as a kind of default prediction when the models lack strong cues.

From a methodological perspective, the German results reinforce two key priorities for future work: refining the feature schema (especially TAML) to capture layered TAM categories more precisely, and probing how predicate type (i.e. Aktionsart) interacts with other features to shape gram choice.

2.3.2.5 English

The RFC experiment achieved an overall accuracy of 93% on the English data, reflecting 139 accurate predictions out of 150 total. The MLP experiment achieved an overall accuracy of 91%, reflecting 137 accurate predictions out of 150 total. English results reveal a high overall performance profile: both models handle most grams with relative ease, with errors concentrated almost entirely in the interaction between past and modal perfects.

2.3.2.5.1 Major Error types

Predicted → Actual	Count	% of All Errors
pastperf → [modal]perf	8	73%

Figure 19: Major English RFC error types, alongside gram occurrence rates in test set

Predicted → Actual	Count	% of All Errors
<i>pastperf</i> → [modal] <i>perf</i>	5	45%

Figure 20: Major English MLP error types, alongside gram occurrence rates in test set

Both models systematically overpredicted *pastperf* in contexts where the target was [modal]*perf*. In the RFC results this accounted for nearly three-quarters of all errors, while the MLP results show a lower proportion but also included two additional *pastperf* → *perf* mispredictions. Together, these errors highlight TAM layering as a key locus of instability in the English data, mirroring difficulties observed in German.

2.3.2.5.2 Feature Metrics

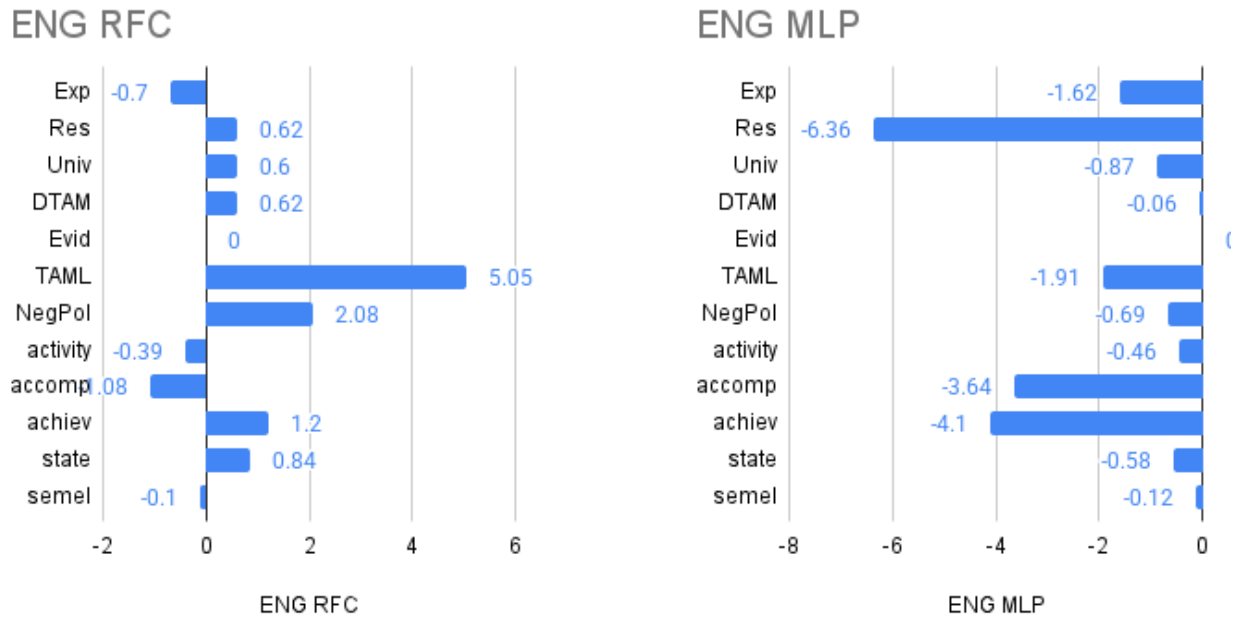


Figure 21: English Weighted Scores (WS) By Feature

	Random Forest (RFC)		MLPClassifier (MLP)	
	FER%	WS	FER%	WS
Experiential	4%	-0.70	4%	-1.62
Resultative	8%	0.62	9%	-6.36
Universal	13%	0.60	13%	-0.87
DTAM	100%	0.62	100%	-0.06
Evidentiality	0%	0.00	0%	0.00
TAML	30%	5.05	33%	-1.91
NegPol	33%	2.08	33%	-0.69
Activity	0%	-0.39	13%	-0.46
Accomplishment	5%	-1.08	5%	-3.64
Achievement	10%	1.20	10%	-4.10
State	20%	0.84	20%	-0.58
Semelfactive	0%	-0.10	0%	-0.12

Figure 22: English Feature Metrics for Random Forests and MLPClassifier

Based on the data presented in the tables above, both models show feature-level differentiation, but their directional bias is inverted. On the one hand, RFC shows broad over-sensitivity, yielding worse-than-baseline performance on most English features, which suggests that the model is “noticing” features but in a way that hurts prediction. This is why so many WS values are positive, i.e. it’s disproportionately making errors on those features, as though it’s over-weighting them or latching onto misleading cues.

Conversely MLP shows broad undersensitivity, performing better than baseline across nearly all features, which suggests that the model is also differentiating features, but almost all WS are negative, meaning it is systematically making fewer errors than baseline; i.e. the model is treating those features as useful guides rather than traps, so performance improves across the board.

The tables above show overall feature performance by classifier; recall that lower WS values indicate stronger performance on that feature. Figure 23 shows the statistical measures for the features deemed most significant by χ^2 analysis, informing the degree to which the feature performance rate impacted overall accuracy.

	p -value	ϕ -value
TAML RFC	0.000	0.468
TAML, MLP	0.000	0.466
Negative Polarity, RFC	0.002	0.294
Negative Polarity, MLP	0.009	0.259

Figure 23: English Statistical Measures

Only two features were deemed statistically significant by way of χ^2 test: TAML and Negative

Polarity. The significance of both features is quite strong, with TAML standing out especially. The TAML ϕ -values also approach the threshold of strong association, though Negative Polarity shows rather weaker association.

To complement the quantitative patterns, Figures 24 and 24 illustrate the one major English error type. While the range of errors is far narrower than in Mandarin or German, the tables highlights the consistent role of the TAML feature in conditioning mispredictions. These patterns provide a bridge between the raw error counts and the broader discussion that follows, clarifying how the models’ difficulties are concentrated in cases of TAM layering.

Error Type	Feature Association(s)	% Error Match
<i>pastperf</i> → <i>[modal]perf</i>	8/8 +TAML, 8/8 modals	100%

Figure 24: English RFC Error Type-Feature Associations

Error Type	Feature Association(s)	% Error Match
<i>pastperf</i> → <i>[modal]perf</i>	5/5 +TAML, 5/5 modals	100%

Figure 25: English MLP Error Type-Feature Associations

Rough interpretation: TAM layering is opaque to the models

2.3.2.5.3 Discussion

Several key insights emerge from the English results. First, the models’ high overall accuracy suggests that the feature schema captures many of the key factors conditioning TAM marking in English. However, the consistent *pastperf* → *[modal]perf* errors indicate that TAM layering remains problematic: both models appear to collapse modal and past perfects into a single category. This reflects a broader limitation of the TAML feature, which conflates distinct phenomena (e.g. tense-shifted versus modal), and is compounded by the use of complex grams in the annotation scheme. As with the German and Mandarin datasets, this points to the need for decomposing both the TAML feature and the morphological complexes into finer-grained categories, enabling future experiments to more precisely evaluate how models handle different layers of featural as well as morphological expression.

Additionally, NegPol in both experiments shows a relatively strong p -value, and a ϕ -value in the weak to nearly-medium strength of association range, though its WS is unexceptional. This suggests that the NegPol feature may be salient to both models, though with only 12/150 tokens in the dataset, the small sample size prevents firm conclusions.

2.4 General Summary & Discussion

The pilot experiment utilized 300 tokens across three languages. While this dataset is not large enough to support major conclusions, it does provide a diagnostic snapshot that informs how

to proceed with the next stage of the project. The features and gram identification are preliminary; the schema is somewhat naïve, and feature assignments and gram identifications were conducted by a single (monolingual English speaker) annotator (myself). These results should therefore be treated as indicative rather than conclusive. Nonetheless, several important lessons emerge for refining both the methodology and the theoretical framing going forward.

Across languages and classifiers, the dominant pattern was not diffuse errors but clustering: a small set of feature contrasts disproportionately accounted for prediction failures. In Mandarin, experiential PLGs were consistently difficult to predict while resultatives were comparatively easy. In German and English, TAM layering (TAML) proved especially challenging, with the German Universal feature was also strongly associated with errors. In both German and Mandarin, Aktionsart distinctions revealed further misalignments: in Mandarin between accomplishments and achievements in overpredictions of *le*, and in German between accomplishments and achievements in resultatives, and between activities and states in universals. In English, overlap between TAML and NegPol appears to have further skewed results for the latter feature.

In this analysis, “difficulty” means that a feature has a higher error rate than expected, as reflected in positive WS values or elevated FER. For PLGs, this indicates that the model struggles to map features consistently onto grams, especially when those features are distributed across multiple grams or expressed in layered ways. It is important to note that the measures reported here are calculated on features rather than grams: the models make predictions at the gram level, but the evaluation abstracts over those predictions to trace how individual features shape performance. This feature-based perspective avoids reifying cross-linguistic categories and instead aligns with a semantic clustering approach, treating features as nodes in a cross-linguistic space whose relative salience and predictability can be compared across languages.

These patterns underscore several areas for methodological refinement going forward, chief among them being decomposition of both the TAML feature and of complex morphological grams across all languages, allowing for more precise evaluation of the components of both meaning and morphological composition. This may in turn drive additional development and/or refinement of the feature schema itself, though such work will not begin until the morphological (and TAML) decomposition provide clearer insights as to how the schema can be improved.

More broadly, these findings reflect a typological question which is currently under discussion in the literature: can perfects/PLGs be treated as a coherent cross-linguistic class? Dahl & Wälchli (2016) argue that perfects are slippery not only across languages but also within them, and the present results echo that difficulty. Similarly, Bertrand et al. (2022) propose decomposing PLGs into a four-way distinction between experientials, resultatives, past perfectives, and “hybrids” (i.e. forms that allow both experiential and resultative readings). This project seeks to address this question by examining cross-linguistic PLGs through a more articulated lens, using an ontological feature schema to explore the semantic space of cross-linguistic PLGs through a more complex (and, hopefully, nuanced) lens.

Future work will focus on three fronts: (i) decomposing complex morphological forms and the TAML feature, (ii) expanding the dataset with more tokens and languages, and (iii) incorporating L1 annotators to improve the reliability of feature coding and gram equivalence judgments. These steps will help ensure that the components of both form and meaning are evaluated on a granular level, that experimental results reflect quantitatively robust and typologically diverse insights, and that ambiguous or edge cases are consistently treated, and will allow the project to move from pilot-scale diagnostics toward a broader typological framework.

3 Dissertation Plan

This dissertation is structured as three interlocking projects, each building on the work of the preceding stages and feeding into a cumulative typological framework. Project 1 refines the methodology and establishes its validity against vector-based distributional baselines. Project 2 applies the refined method to address core linguistic questions about the semantics and typology of PLGs. Project 3 then tests the scalability of the approach, evaluating whether it can generalize to novel tokens and languages. Together, these projects move from methodological refinement, to linguistic analysis, to typological generalization, consolidating the thesis’s contribution at the intersection of computational modeling and linguistic theory and typology.

3.1 Strucrure of the Dissertation

3.1.1 Project 1: Methodological refinement and model comparison

The first project builds directly on the pilot experiment, scaling it into a fuller test of the ontology-based methodology. Methodological refinements will include decomposing the coarse-grained TAML feature into more precise categories (e.g. tense-shifted versus modal perfects), and separating out multiple layers of morphological expression such as mood distinctions (e.g. Indikativ, Konjunktiv I & II, individual adverbs/particles, etc.). L1 speakers will be recruited as annotators to identify grams in the non-English data and flag potential meaning components that are expressed there but not in the corresponding English; top-level feature annotation will remain under my domain. The dataset will expand to a minimum of 500 and ideally up to 1,000 tokens, across English and 6-7 typologically diverse languages (including Mandarin and German), with the potential incorporation of additional corpora such as TED Talks (TBD). To capture contrasts missed in English, the “reverse seeding” method (as described in the pilot experiment’s methodology section, 2.2; (Dahl & Wälchli, 2016)) will also be explored, identifying PLGs in the non-English texts corresponding to non-PLG expressions in English. Finally, the ontology-based feature schema will be compared (and iteratively updated) against vector-based distributional approaches, specifically BERT and S-BERT, to evaluate whether it provides interpretability and typological insight beyond what those models capture.

3.1.2 Project 2: Core linguistic analysis

The second project shifts focus from methodology to linguistic interpretation and insights. Using the refined feature schema and expanded dataset, I will investigate what PLGs reveal about cross-linguistic TAM semantics, as viewed through the lens of PLGs. Analytical tools such as principal component analysis (PCA) will be used to map the semantic space occupied by these grams, while results from the classifiers will provide evidence about which contrasts are most salient across languages. This analysis will ask whether PLGs can be treated as a coherent cross-linguistic category, or whether they fragment along dimensions such as morphological type (e.g. affixal versus periphrastic versus aspectual/adverbial marking), TAM system (the number and kind of distinctions overtly marked in a given language), and meaning components (feature constellations/profiles). Data from all other languages in the study will be compared to English in terms of both morphosyntactic strategies and semantic contrasts. This project thus grounds the thesis in linguistic theory, using the experimental methodology as a tool to probe fundamental questions about the nature of PLGs and their place in cross-linguistic typology.

3.1.3 Project 3: Generalization and scalability

The third project tests the broader impact of the methodology: can the ontology-based schema scale to new domains and unseen data? Building on results from the first two projects, I will evaluate whether this feature-based framework can generalize to novel tokens and languages, focusing in particular on the DoReCo corpora as a source of new data. This stage addresses the central question of whether the schema can serve as a typologically meaningful and computationally robust framework for modeling PLGs beyond the languages and texts included in training. Specific tests will evaluate performance on unseen grams and on typologically distinct languages, probing both accuracy and interpretability. This project represents the culminating stage of the thesis: if successful, it will demonstrate that the methodology is not only valid for analyzing PLGs in controlled settings, but also scalable and applicable in ways that advance typological research and support cross-linguistic generalizations.

3.2 Planning & Strategy

3.2.1 Potential Challenges & Mitigation

The most immediate challenge is annotation, which is both time-intensive and prone to subjectivity. To address this, the project will recruit and train L1 research assistants, whose contributions will improve both speed and reliability of annotation. Guidelines will be iteratively refined to ensure consistency, even in ambiguous cases, and an inter-annotator agreement (IAI) will be employed to further assure that data annotation is consistent across annotators. A broader risk is that the methodology may ultimately fail to produce strong or consistent results. Even in that scenario, however, the findings would remain informative: they would highlight the limits of current feature-based approaches and clarify which aspects of PLGs resist formalization, offering valuable insight for both typology and NLP.

3.2.2 Dissemination

Each project is designed to yield conference presentations and eventual publications.

- Project 1 (methodological refinement and model comparison): CoNLL and a corpus-linguistic journal
- Project 2 (core linguistic analysis): Typology at the Crossroads
- Project 3 (generalization and scalability): Possible venues include LREC (Language Resources and Evaluation Conference), COLING, CoNLL, or SIGTYP

3.2.3 Timeline

The project will proceed on a compressed schedule to allow for defense by August 2026.

- October 2025 - January 2026: Complete methodological refinement and run model comparison experiments (Chapter 1) and submit initial conference paper
- February - May 2026: Craft core linguistic analysis chapter; prepare submissions to CoNLL/Typology at the Crossroads
- June - September 2026: Conduct generalization/scalability experiments (Chapter 3); prepare submission to LREC or ACL-related venue

- Fall 2026: Craft bookend chapters and finalize dissertation
- December 2026: Dissertation defense

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